Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V) MERN

Practical II

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Class: TYIT	Batch:1
Date of Assignment: 30-06-2026	Date/Time of Submission: 30-06-2026

2. Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

CODE:

```
function calculateSalary name   basicSalary
let bonous = 5000
let totalSalary = basicSalary + bonous
console log "Emplyoee Name: " name
console log "Basic Salary: "  basicSalary
console log "Bonous : "  bonous
console log "Total Salry: " totalSalary

calculateSalary "Amit" 30000
console log "Roshni"
```

OUTPUT:

```
PS C:\MERN PRACTICAL> node prac2a.js
Emplyoee Name:  Amit
Basic Salary:  30000
Bonous :  5000
Total Salry:  35000
Roshni
```

CODE:

```
function genbill product  quantity  price
let total = quantity * price
let totalbill = total + total*0.18
console log "Product Name: " product
console log "Quantity "  quantity
```

```
console log "Price per Item: " price
console log "Total Bill: " totalbill

genbill "Mobile" 2 8000
console log "Roshni"
```

OUTPUT:

```
PS C:\MERN PRACTICAL> node prac2a.js
Product Name:  Mobile
Quantity  2
Price per Item:  8000
Total Bill:  16000
Roshni
```

b. Work with arrays and objects in JavaScript **CODE:**

```
let fruits =  "Mango" "Apple" "Pear"
let student =
 name: "Roshni"
 age : 20
 city: "Mumbai"

console log "Fruits: " fruits
console log "Second Fruit: " fruits 1
console log "Student Name: " student name
console log "Student Age: " student age
console log "Student City: " student city
```

OUTPUT:

```
PS C:\MERN PRACTICAL> node prac2b.js
Fruits:  [ 'Mango', 'Apple', 'Pear' ]
Second Fruit:  Apple
Student Name:  Roshni
Student Age:  20
Student City:  Mumbai
```

. c. Demonstrate use of arrow functions.

CODE:

```
let add = a b  =>
 return a + b

console log "SUM = " add 20 10
let sub = a b  =>
```

```
 return a - b

console log "SUB = " sub 20 10
let mul = a b  =>
 return a * b

console log "MUL = " mul 20 10
console log "Roshni"
```

OUTPUT:

```
PS C:\MERN PRACTICAL> node prac2c.js
SUM =  30
SUB =  10
MUL =  200
Roshni
```

d. Write a program using array methods (map, filter, reduce).

CODE:

```
let numbers =  10 20 30 40 50
let doubled = numbers map num => num*2   let
greater25 = numbers filter num => num >25   let total
= numbers reduce  sum num  => sum + num 0

console log "Original Array: " numbers
console log "Map: " doubled
console log "Filter: " greater25
console log "Reduce: " total
console log "Roshni"
```

OUTPUT:

```
PS C:\MERN PRACTICAL> node prac2d.js
Original Array:  [ 10, 20, 30, 40, 50 ]
Map:  [ 20, 40, 60, 80, 100 ]
Filter:  [ 30, 40, 50 ]
Reduce:  150
Roshni
```

e. Create a program that manipulates objects and displays output.

CODE:

```
let student =
 rollno: 41
 name: "Roshni"
```

```
 marks: 90

console log "Original Object: "
console log student
console log "\n Accessing Properties"
console log "Name: " student name
console log "Marks: "  student marks

student marks = 95
console log "\n After Updating Marks"
console log student

student city = "Mumbai"
console log "\n After Adding City"
console log student
console log "Roshni"
```

OUTPUT:

```
PS C:\MERN PRACTICAL> node prac2e.js
Original Object:
{ rollno: 41, name: 'Roshni', marks: 90 }

 Accessing Properties
Name:  Roshni
Marks:  90

 After Updating Marks
{ rollno: 41, name: 'Roshni', marks: 95 }

 After Adding City
{ rollno: 41, name: 'Roshni', marks: 95, city: 'Mumbai' }
Roshni
```

f. Implement a small JavaScript program combining functions, arrays, and objects.

CODE:

```
let students =


 rollNo: 101
 name: "Rahul"
 marks: 85
```

```
 rollNo: 102
 name: "Varun"
 marks: 90


 rollNo: 103
 name: "Amit"
 marks: 95



function display s

 console log "Student Details"
 console log "---------------------------"
for  let stud of s

 console log `Roll No : ${stud rollNo}`
console log `Name : ${stud name}`
console log `Marks: ${stud marks}`
console log "---------------------------"

display students
console log "Roshni"
```

OUTPUT:

```
PS C:\MERN PRACTICAL> node prac2f.js
Student Details
---------------------------
Roll No : 101
Name : Rahul
Marks: 85
---------------------------
Roll No : 102
Name : Varun
Marks: 90
---------------------------
Roll No : 103
Name : Amit
Marks: 95
---------------------------
Roshni
```